

Methodology

Universe. Fourteen US-listed iShares ETFs, USD, total return (dividends reinvested, net of fund fees). Equity: IVV (S&P 500), IEFA (developed ex-US), IEMG (emerging), QUAL and VLUE (MSCI USA Quality and Value factor). Bonds: AGG, TIP, IEF, TLT, SHY, LQD, HYG, EMB. Real assets: IAU (gold). Cash: SHV. Prices, Treasury yields, breakevens and CPI come from Yahoo Finance and FRED; holdings, sectors, countries, currencies and durations come from the iShares holdings files. All of it lands in one SQLite database.

Window. Daily data from 2003; the backtest starts 2013-08-01, the first full month after QUAL listed, so all four portfolios run on the same days. It covers 2015-16, Q4 2018, COVID, 2022, 2023 and April 2025. The 2008 crisis sits before the window and is handled by stress replay with proxies, not by the backtest.

Portfolios.

Portfolio	Construction	Rebalancing
Benchmark 60/40	60% ACWI, 40% AGG	monthly, no costs
Strategic 60/40	IVV 36, IEFA 16, IEMG 8, AGG 32, TIP 8	quarterly check, trade only if a sleeve is >2.5pp from target
Minimum Variance	long-only min-variance over 11 ETFs, Ledoit-Wolf shrunk covariance on 252 days, max 25% per ETF, equities >= 20%	monthly re-optimisation, no trade if the new target is within 1pp
Quality-Value	the strategic portfolio with its 36% US equity sleeve split 50/50 QUAL/VLUE	same as strategic

The strategic and quality-value portfolios differ **only** in the US equity sleeve, so the gap between them is a factor effect and nothing else.

Backtest mechanics. Weights drift daily with returns. Targets are computed from data up to a signal date and executed at the next close (one-day lag, no look-ahead). Trading costs are one-way basis points per ETF (1bp IVV to 5bp VLUE, covering half-spread and impact), charged on the weight actually traded. Turnover is one-way: half the sum of absolute weight changes. Fund fees are already inside total returns. Cash return is SHV; it is also the risk-free rate in Sharpe ratios.

Risk. Volatility, Sharpe, Sortino, maximum drawdown and recovery, tracking error and information ratio versus the benchmark, beta, up/down capture, historical VaR and CVaR at 95%. Ex-ante risk contributions use a Ledoit-Wolf covariance matrix and Euler decomposition, so each holding's contribution sums to portfolio volatility.

Exposures. Look-through: portfolio weight $\times$ each fund's holdings gives sector (GICS for equity, issuer type for bonds), country of risk and market currency. Duration comes from the per-bond durations in the iShares files, weighted up to the fund and then to the portfolio; TIPS duration is treated as real duration and reported separately. Spread duration is duration $\times$ the fund's credit share.

Attribution. Brinson-Fachler by segment, daily, Carino-linked: allocation is (portfolio weight – benchmark weight) $\times$ (segment benchmark return – total benchmark return); selection is benchmark weight $\times$ (segment portfolio return – segment benchmark return); interaction is reported with selection. Segment benchmark returns use a reference ETF per segment (IVV, IEFA, IEMG, AGG, TIP, IEF, LQD, IAU, SHV). The benchmark's own regional mix is estimated monthly by returns-based style analysis (Sharpe 1992) of ACWI on IVV/IEFA/IEMG, so allocation can be measured region by region. Two terms close the identity exactly: trading costs, and a benchmark residual (ACWI/AGG versus the composite, fees, small caps, Canada). Effects sum to the active return every single day before linking.

Factor model. Weekly OLS of excess (and active) returns on ten ETF-built factors: equity (ACWI–SHV), size (IWM–IVV), value (IWD–IWF), quality (QUAL–IVV), momentum (MTUM–IVV), rates (IEF–SHV), credit (LQD–IEF), inflation (TIP–IEF), dollar (UUP), gold (IAU–SHV). Return contribution is beta $\times$ the factor's realised premium; risk shares are the Euler decomposition of regression variance, with the residual shown as specific risk.

Stress tests. Two kinds. *Historical replay* holds today's weights through nine past episodes; ETFs that did not exist are proxied (IEFA→EFA, IEMG→EEM, QUAL→IVV, VLUE→IWD, ACWI→IVV/EFA/EEM, EMB→LQD/HYG) and the proxy share is reported beside the result. *Hypothetical shocks* are instantaneous: bonds by $-D \cdot \Delta y$ plus convexity plus spread duration $\times$ spread change (TIPS on real yields, $\Delta_{\text{real}} = \Delta_{\text{nominal}} - \Delta_{\text{breakeven}}$); equities by their 3-year weekly betas to ACWI and to the dollar; gold by assumption. Scenarios that specify no equity move imply one from ACWI's own regression on 10y yield changes and the dollar.

What this is not. A backtest of fixed rules on ETF price history, not live performance. It ignores taxes, securities lending, bid-ask variation through time, cash drag and capacity, and it uses one holdings snapshot for look-through exposures. Every ETF here still exists, so survivorship is not an issue, but the factor ETFs were chosen with hindsight about which factors are interesting.